

Joint Estimation of Wideband DOAs and Phase Errors for Distributed Array via Sparse Bayesian Inference

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Abstract—This paper addresses the joint estimation of wideband DOAs and phase errors in a distributed array system with unknown phase errors. In order to capture the coherent gain introduced by the distributed array, the joint estimation problem is formulated as a sparse Bayesian inference (SBI) problem with uncertain parameters. We propose an expectation maximization (EM) based subspace-constrained sparse Bayesian inference framework to efficiently solve this problem, where the E-step performs Bayesian estimation of the sparse signal, while the M-step updates the dynamic grid parameters to achieve high-resolution DOA estimation and compensate the phase error to improve the coherent gain of the distributed array. To reduce the computational complexity, we employ the subspace-constrained variational Bayesian inference (SC-VBI) method to avoid the high-dimensional matrix inverse in the E-step. Numerical simulations demonstrate the superior performance of the proposed algorithm compared to other baseline methods.

Index Terms—Direction-of-arrival, distributed array, phase errors, variational Bayesian inference.

I. INTRODUCTION

Wideband Direction-of-Arrival (DOA) estimation is pivotal in array signal processing. Recently, distributed arrays have gained attention due to their higher spatial diversity and spectral efficiency compared to monolithic arrays [1]. However, distributed arrays face a critical challenge: phase desynchronization between subarrays caused by independent local oscillators and position uncertainties [2]. While frequency synchronization is feasible via hardware (e.g., PLLs) [3], eliminating residual phase errors remains difficult. Uncalibrated phase errors destroy coherent gain, severely degrading estimation performance.

Few studies have explored wideband DOA estimation for distributed arrays in the presence of phase errors. Conventional wideband DOA estimation algorithms, such as subspace-based methods, rely on a sufficient number of snapshots and sensors and are susceptible to aliasing effects [4]. The work in [5], [6] introduces a sparse Bayesian learning (SBL) framework that leverages a structured prior design to effectively handle

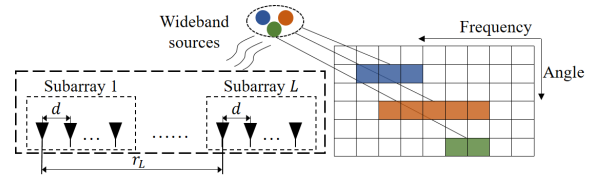


Fig. 1. The considered distributed array system.

signals with aliased or partially overlapping frequency bands. However, this method presupposes a perfectly calibrated array and is not directly applicable to scenarios with uncalibrated phase errors. [7] addresses the sparse recovery problem with phase errors within the SBL framework, but it is limited to the narrowband signal assumption. In summary, conventional subspace-based wideband DOA estimation methods are unsuitable for sources with partially overlapping frequency bands, while existing SBL-based approaches lack a design for the joint estimation of wideband DOAs, sparse signals, and array imperfections. Furthermore, they often exhibit high computational complexity due to the matrix inverse, especially when the number of sensors and angular grid points is large.

In this paper, we propose a novel scheme for the joint estimation of wideband DOAs, phase errors, and sparse signals in a distributed array system with unknown phase errors. We formulate this as a sparse Bayesian inference (SBI) problem with uncertain parameters and introduce a solution that integrates subspace-constrained variational Bayesian inference (SC-VBI) within an expectation maximization (EM) framework. The proposed algorithm iteratively alternates between two main steps, where E-step performs sparse signals recovery via SC-VBI, while M-step updates the dynamic grid parameters to achieve high-resolution DOA estimation and compensate the phase errors to improve the coherent gain of the distributed array.

II. SYSTEM MODEL

Consider a distributed array system with L collinear subarrays. As depicted in Fig. 1, each subarray is an M -element uniform linear array (ULA) with spacing d . Let the first sensor

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of the first subarray be the reference. The position of the l -th subarray is denoted by r_l . The array intercepts K far-field wideband sources from angles $\theta = [\theta_1, \theta_2, \dots, \theta_K]^T$, which may occupy distinct or partially overlapping frequency bands. The frequency-domain received signal at the l -th subarray for the n -th frequency bin f_n is [6]:

$$\mathbf{y}_{l,n} = \sum_{k=1}^K e^{j\varphi_l} x_{n,k} e^{-j2\pi f_n \Delta\tau_l(\theta_k)} \mathbf{a}_n(\theta_k) + \mathbf{n}_{l,n}, \quad (1)$$

where $e^{j\varphi_l}$ is the unknown phase error of the l -th subarray (constant over the observation), $x_{n,k}$ denotes the signal coefficient, $\Delta\tau_l(\theta_k) = r_l \sin \theta_k / c$ represents the relative propagation delay. The noise $\mathbf{n}_{l,n} \sim \mathcal{CN}(\mathbf{n}_{l,n}; 0, 1/\gamma \mathbf{I}_M)$ is the additive white Gaussian noise (AWGN) with variance $1/\gamma$. We use a Gamma distribution with shape parameter c and rate parameter d to model the noise precision, i.e., $p(\gamma) = \Gamma(\gamma; c, d)$, and it can be learned automatically by the algorithm. The array response vector is $\mathbf{a}_n(\theta_k) = [1, e^{-\frac{j2\pi f_n d \sin \theta_k}{c}}, \dots, e^{-\frac{j2\pi f_n (M-1)d \sin \theta_k}{c}}]^T$.

Note that if the k -th source does not occupy the n -th frequency component, we have $x_{n,k} = 0$. Moreover, we assume non-cooperative sources, i.e., both the frequency band occupied by each source and the frequency domain signals $x_{n,k}$'s are unknown.

III. PROBLEM FORMULATION

A. Sparse Representation of Frequency Domain Signal

We discretize the angular space into a dynamic grid $\Theta = [\theta_1, \dots, \theta_Q]^T$, which is uniformly initialized and iteratively refined. The off-grid effect arises because the true DOAs are continuous-valued and generally do not coincide exactly with the initial discretized grid points. This basis mismatch causes energy leakage over neighboring atoms and leads to biased angle estimation. By treating the grid points in Θ as unknown parameters and refining them during the M-step, the proposed method alleviates the off-grid effect and improves the angular resolution. Accordingly, the sparse basis in the angular domain is given by $\mathbf{A}_n(\Theta) = [\mathbf{a}_n(\theta_1), \dots, \mathbf{a}_n(\theta_Q)]$. The imperfection matrix incorporating delays and phase errors is $\mathbf{D}_{l,n}(\Theta, \varphi_l) = \text{Diag}[e^{j(\varphi_l - 2\pi f_n \Delta\tau_l(\theta_1))}, \dots, e^{j(\varphi_l - 2\pi f_n \Delta\tau_l(\theta_Q))}]$. Then the angular-domain sparse representation of the received signal for the l -th subarray and the n -th frequency component is given by

$$\mathbf{y}_{l,n} = \mathbf{F}_{l,n}(\xi) \mathbf{x}_n + \mathbf{n}_{l,n}, \quad (2)$$

where $\mathbf{F}_{l,n}(\xi) \triangleq \mathbf{A}_n(\Theta) \mathbf{D}_{l,n}(\Theta, \varphi_l)$ denotes the observation matrix, $\xi \triangleq \{\Theta, \varphi\}$ denotes uncertain parameters, $\mathbf{x}_n = [x_{n,1}, \dots, x_{n,Q}]^T \in \mathbb{C}^{Q \times 1}$ denotes sparse frequency signal vector which contains at most K non-zero elements corresponding to K sources and the grid positions of the non-zero elements correspond to the incident angles.

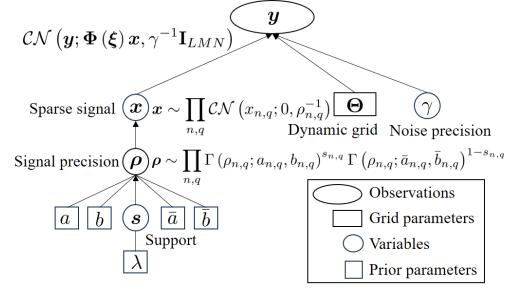


Fig. 2. Illustration of the hierarchical sparse prior model.

B. Three-Layer Hierarchical Sparse Prior Model

We employ a three-layer sparse prior model to characterize the structured sparsity of $\mathbf{x}_n$'s as illustrated in Fig. 2. A binary support vector $\mathbf{s}_n \triangleq [s_{n,1}, \dots, s_{n,Q}]^T \in \{0, 1\}^Q$ is introduced to indicate whether $x_{n,q}$ is active ($s_{n,q} = 1$). We use a Markov chain to model the burst sparsity in the frequency domain: $p(\mathbf{s}) = \prod_q p(s_{1,q}) \prod_n p(s_{n,q} | s_{n-1,q})$, characterized by transition probabilities p_{01} and p_{10} . Conditioned on $\mathbf{s}$ the signal precision ρ follows a Gamma distribution: $p(\rho | \mathbf{s}) = \prod_{n,q} \Gamma(\rho_{n,q}; a_{n,q}, b_{n,q})^{s_{n,q}} \Gamma(\rho_{n,q}; \bar{a}_{n,q}, \bar{b}_{n,q})^{1-s_{n,q}}$. Parameters are chosen such that active signals have low precision (high variance) and inactive ones have high precision (near-zero variance). Conditioned on ρ the signal follows a Gauss distribution: $p(\mathbf{x} | \rho) = \prod_{n,q} \mathcal{CN}(x_{n,q}; 0, \rho_{n,q}^{-1})$. Then the joint distribution of $\mathbf{x}$, ρ , and $\mathbf{s}$ can be expressed as

$$p(\mathbf{x}, \rho, \mathbf{s}) = \underbrace{p(\mathbf{s})}_{\text{Support}} \underbrace{p(\rho | \mathbf{s})}_{\text{Precision}} \underbrace{p(\mathbf{x} | \rho)}_{\text{Sparse signal}}. \quad (3)$$

C. Sparse Bayesian Inference with Uncertain Parameters

Using the angular-domain sparse representation in (2), the global signal model across all subarrays is cast as a linear inverse problem with uncertain parameters:

$$\mathbf{y} = \Phi(\xi) \mathbf{x} + \mathbf{n}, \quad (4)$$

where $\Phi(\xi) = \text{BlockDiag}(\Phi_1, \dots, \Phi_N)$ is the sensing matrix dependent on ξ and $\Phi_n = [\mathbf{F}_{1,n}(\xi)^T, \dots, \mathbf{F}_{L,n}(\xi)^T]^T$. Our objective is to jointly estimate the sparse signal $\mathbf{x}$ and the parameters ξ . The parameters are estimated via the maximum a posterior (MAP) estimator $\xi^* = \arg\max_{\xi} \ln \int_{\mathbf{v}} p(\mathbf{y}, \mathbf{v}; \xi) p(\xi)$,

where $\mathbf{v} \triangleq \{\mathbf{x}, \rho, \mathbf{s}, \gamma\}$ is the collection of all variables, $p(\xi)$ denotes the prior distribution of the sensing parameters, which is usually assumed to be uniform distribution in the absence of specific prior knowledge. Once we obtain the MAP estimate of ξ^* , we can update the conditional marginal posterior for $\mathbf{v}$ via SC-VBI given the uncertain parameters ξ^* .

The DOAs θ can be obtained by searching in the estimated support vector $\hat{\mathbf{s}}$. Let $\hat{\mathcal{S}} = \{\hat{q} | \sum_n \hat{s}_{n,q} \neq 0\}$ be the set of indices with significant energy, the estimated DOAs correspond to the grid points in the dynamic grid Θ indexed by $\hat{\mathcal{S}}$:

$$\hat{\theta}_k = \Theta_{\hat{q}}, \hat{q} \in \hat{\mathcal{S}}, k = 1, \dots, |\hat{\mathcal{S}}|. \quad (5)$$

IV. PROPOSED JE-DPS ALGORITHM

The proposed joint estimation of DOAs, phase errors, and sparse signal (JE-DPS) algorithm iteratively performs the E-step and M-step as illustrated in Fig. 3.

A. JE-DPS-E Step (SC-VBI)

The JE-DPS-E step consists of two modules: Module A and Module B. In particular, Module A adopts SC-VBI to compute the approximate posterior based on the independent sparse prior information from Module B. Module B adopts sum-product message passing [8] to compute the marginal posterior based on the decoupled observation information from Module A. Module A and B exchange extrinsic messages until convergence.

Given the uncertain parameter ξ , we have the following effective prior distribution

$$\hat{p}(\mathbf{v}) = p(\gamma) \hat{p}(\mathbf{s}) p(\boldsymbol{\rho} | \mathbf{s}) p(\mathbf{x} | \boldsymbol{\rho}), \quad (6)$$

where $\hat{p}(\mathbf{s}) = \prod_{n,q} \phi_{A,n,q}(s_{n,q})$ is independent prior from Module B. For convenience, we use $v^i, i \in \mathcal{H} \triangleq \{i | \forall \mathbf{v}^i \in \mathbf{v}\}$ to denote an individual variable in $\mathbf{v}$. Denote $\hat{p}(\mathbf{v} | \mathbf{y})$ as the posterior distribution of $\mathbf{v}$ with the effective prior $\hat{p}(\mathbf{v})$ in (6). Based on the VBI method [9], the approximate conditional marginal posterior could be calculated by minimizing the Kullback-Leibler divergence (KLD) between $\hat{p}(\mathbf{v} | \mathbf{y})$ and $q(\mathbf{v})$, subject to a factorized form constraint as

$$\begin{aligned} \mathcal{A}_{\text{VBI}} : q^*(\mathbf{v}) &= \underset{q(\mathbf{v})}{\operatorname{argmin}} \int q(\mathbf{v}) \ln \frac{q(\mathbf{v})}{\hat{p}(\mathbf{v} | \mathbf{y})} d\mathbf{v}, \\ \text{s.t. } q(\mathbf{v}) &= \prod_{i \in \mathcal{H}} q(v^i), \int q(v^i) dv^i = 1, \\ q(\mathbf{x}) &= \mathcal{CN}(\mathbf{x}; \boldsymbol{\mu}, \operatorname{Diag}(\boldsymbol{\sigma}^2)). \end{aligned} \quad (7)$$

We add an additional constraint that $q(\mathbf{x})$ is a Gaussian distribution with mean $\boldsymbol{\mu}$ and diagonal covariance matrix $\operatorname{Diag}(\boldsymbol{\sigma}^2)$. Such a constraint helps to reduce the complexity of calculating the posterior covariance, but the posterior mean still requires a high-dimensional matrix inverse, whose complexity can be further reduced using a subspace-constrained method. A stationary solution demonstrated in [10] for Problem $\mathcal{A}_{\text{VBI}}$ is given by

$$q(v^i) \propto \exp \left(\langle \ln \hat{p}(\mathbf{v}, \mathbf{y}) \rangle_{\prod_{l \neq i} q(v^l)} \right), \quad (8)$$

where $\langle f(x) \rangle_{q(x)} = \int f(x) q(x) dx$. Based on the joint distribution (6) and the stationary solution of $\mathcal{A}_{\text{VBI}}$ (8), each optimal variational distribution $q(v^i)$ can be derived.

However, when $v^i = \mathbf{x}$, the optimal $q(\mathbf{x})$ is no longer given by (8) due to the independence constraint. In [11], it is proven that for given $q(v^i)$, $v^i \neq \mathbf{x}$, the optimal posterior mean $\boldsymbol{\mu}$ and variance vector $\boldsymbol{\sigma}^2$ that minimizes the KLD subject to $q(\mathbf{x}) = \mathcal{CN}(\mathbf{x}; \boldsymbol{\mu}, \operatorname{Diag}(\boldsymbol{\sigma}^2))$ are given by

$$\begin{aligned} \boldsymbol{\mu} &= \underset{\mathbf{u}}{\operatorname{argmin}} \varphi(\mathbf{u}) \triangleq \mathbf{u}^H \mathbf{W} \mathbf{u} - 2\Re \{ \mathbf{u}^H \mathbf{b} \} \\ &= \mathbf{W}^{-1} \mathbf{b}, \\ \boldsymbol{\sigma}^2 &= \left[W_{1,1}^{-1}, \dots, W_{1,Q}^{-1}, \dots, W_{N,1}^{-1}, \dots, W_{N,Q}^{-1} \right]^T, \end{aligned} \quad (9)$$

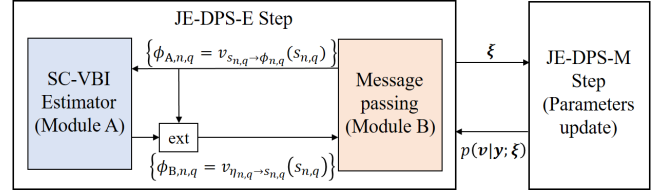


Fig. 3. The proposed JE-DPS algorithm.

where $\mathbf{W} = \langle \gamma \rangle \boldsymbol{\Phi}(\boldsymbol{\xi})^H \boldsymbol{\Phi}(\boldsymbol{\xi}) + \operatorname{Diag}(\langle \boldsymbol{\rho} \rangle)$, $W_{n,q}$ is the $(n-1)Q + q$ -th diagonal element of $\mathbf{W}$, and $\mathbf{b} = \langle \gamma \rangle \boldsymbol{\Phi}(\boldsymbol{\xi})^H \mathbf{y}$.

To avoid the computationally prohibitive high-dimensional matrix inversion $\boldsymbol{\mu} = \mathbf{W}^{-1} \mathbf{b}$, we efficiently solve $\boldsymbol{\mu} = \operatorname{argmin}_{\boldsymbol{\mu}} \varphi(\mathbf{u})$ via a gradient method initialized by a low-complexity subspace solution. Specifically, let $\hat{\mathcal{S}}$ denotes the estimated support of $\mathbf{x}$ obtained in the previous iteration. Define a vector $\boldsymbol{\mu}^{\mathcal{S}}$ such that

$$\boldsymbol{\mu}_{\hat{\mathcal{S}}}^{\mathcal{S}} = \langle \gamma \rangle \mathbf{W}_{\hat{\mathcal{S}}}^{-1} \boldsymbol{\Phi}(\boldsymbol{\xi})_{\hat{\mathcal{S}}}^H \mathbf{y}, \quad (10)$$

and $\boldsymbol{\mu}_{\hat{\mathcal{S}}^c}^{\mathcal{S}} = \mathbf{0}$, where $\boldsymbol{\mu}_{\hat{\mathcal{S}}}^{\mathcal{S}} \in \mathbb{C}^{|\hat{\mathcal{S}}| \times 1}$, $\mathbf{W}_{\hat{\mathcal{S}}} \in \mathbb{C}^{|\hat{\mathcal{S}}| \times |\hat{\mathcal{S}}|}$, and $\boldsymbol{\Phi}(\boldsymbol{\xi})_{\hat{\mathcal{S}}} \in \mathbb{C}^{LNM \times |\hat{\mathcal{S}}|}$ are the subvector/submatrices of $\boldsymbol{\mu}^{\mathcal{S}}$, $\mathbf{W}$, $\boldsymbol{\Phi}(\boldsymbol{\xi})$ with the element/column/row indices lying in $\hat{\mathcal{S}}$, and $\boldsymbol{\mu}_{\hat{\mathcal{S}}^c}^{\mathcal{S}}$ contains the rest elements of $\boldsymbol{\mu}^{\mathcal{S}}$. If $\hat{\mathcal{S}} = \mathcal{S}$, $\boldsymbol{\mu}^{\mathcal{S}}$ is expected to provide a good initial solution for minimizing $\varphi(\boldsymbol{\mu})$ in (9) based on the gradient method.

To enhance robustness against support errors, the gradient search is initialized at $\boldsymbol{\mu}^0$, chosen as either $\boldsymbol{\mu}^{\mathcal{S}}$ or the previous estimate $\boldsymbol{\mu}^p$, whichever yields a lower cost $\varphi(\cdot)$. Starting from $\boldsymbol{\mu}^0$, we apply the gradient update for $B \geq 1$ times. Specifically, in the i -th iteration, the posterior mean is updated as

$$\boldsymbol{\mu}^{(i)} = \boldsymbol{\mu}^{(i-1)} - \epsilon^{(i)} \nabla_{\mathbf{u}} \varphi(\mathbf{u}) \Big|_{\mathbf{u}=\boldsymbol{\mu}^{(i-1)}}, \quad (11)$$

where $\epsilon^{(i)}$ is the step size determined by the Armijo rule.

Such a robust design ensures that the objective value $\varphi(\mathbf{u})$ is decreased monotonically in each iteration until convergence to a stationary solution as proved in [11]. Once we obtain the estimated posterior mean $\boldsymbol{\mu}$, we can obtain an estimated support $\hat{\mathcal{S}}$ for the next iteration by finding the elements in $\boldsymbol{\mu}$ with sufficiently large energy. In the first iteration, we can simplify set $\hat{\mathcal{S}}$ according to available prior information in practice or a simple baseline algorithm (e.g., using OMP). The update equations of remaining variables $\boldsymbol{\rho}, \mathbf{s}, \gamma$ are the same as the Turbo-VBI algorithm proposed in [10], and are omitted for conciseness.

After that, the extrinsic message $\phi_{B,n,q} \triangleq v_{\eta_{n,q} \rightarrow s_{n,q}}(s_{n,q})$ from Module A to Module B can be calculated from the posterior $q(s_{n,q})$ by subtracting the input message $\phi_{A,n,q} \triangleq v_{s_{n,q} \rightarrow \phi_{B,n,q}}(s_{n,q})$. Then in Module B, sum product message passing (SPMP) similar to [10] is performed over $\mathcal{G}_B$ to calculate the message $v_{s_{n,q} \rightarrow \phi_{B,n,q}}(s_{n,q})$.

The complexity reduction of SC-VBI mainly comes from avoiding the direct inversion of the full matrix $\mathbf{W} \in \mathbb{C}^{NQ \times NQ}$ in (9). In IC-VBI, the dominant cost is the inversion of $\mathbf{W}$,

which scales cubically with NQ . By contrast, SC-VBI only inverts the reduced matrix $\mathbf{W}_{\hat{\mathcal{S}}} \in \mathbb{C}^{|\hat{\mathcal{S}}| \times |\hat{\mathcal{S}}|}$ in (10), followed by a few gradient updates for posterior mean refinement. Therefore, when $|\hat{\mathcal{S}}| \ll NQ$, SC-VBI substantially reduces the computational burden while retaining nearly the same estimation accuracy as IC-VBI.

The above algorithm for the E-step is called SC-VBI, which can achieve comparable performance as the original Turbo-VBI algorithm but with much lower complexity.

B. JE-DPS-M Step

This step updates $\xi = \{\Theta, \varphi\}$ by maximizing the expected log-likelihood. Since $\ln p(\mathbf{y}, \xi)$ is intractable, we maximize the surrogate function constructed via the EM method:

$$\begin{aligned} Q(\xi; \xi^{(t)}) &= \int_{\mathbf{v}} q^{(t)}(\mathbf{v}) \ln \frac{p(\mathbf{y}, \mathbf{v}; \xi)}{q^{(t)}(\mathbf{v})} + \ln p(\xi) \\ &= -\langle \gamma \rangle \left[\|\mathbf{y} - \Phi(\xi) \mu\|^2 + \right. \\ &\quad \left. \text{tr}(\Phi(\xi) \text{Diag}(\sigma^2) \Phi(\xi)^H) \right] + \ln p(\xi) + C, \end{aligned} \quad (12)$$

where C is a constant. When $q^{(t)}(\mathbf{v}) = p(\mathbf{v} | \mathbf{y}; \xi^{(t)})$, it can be shown that the gradient of $Q(\xi; \xi^{(t)})$ is consistent with $\ln p(\mathbf{y}, \xi)$ [10], and thus we can apply the gradient ascent method to update ξ as $\Theta^{(t+1)} = \Theta^{(t)} + \varepsilon_{\Theta}^{(t)} \frac{\partial Q(\Theta^{(t)}, \varphi^{(t)}; \xi^{(t)})}{\partial \Theta}$ and $\varphi^{(t+1)} = \varphi^{(t)} + \varepsilon_{\varphi}^{(t)} \frac{\partial Q(\Theta^{(t+1)}, \varphi^{(t)}; \xi^{(t)})}{\partial \varphi}$, where $\varepsilon_{\Theta}^{(t)}$ and $\varepsilon_{\varphi}^{(t)}$ are the step size. Gradient ascent is used to update Θ and φ . The outer iterations stop when the relative change of $Q(\xi; \xi^{(t)})$ is below a threshold or when the maximum iteration number is reached.

V. SIMULATION

We evaluate the proposed algorithm and several baseline methods using a simulated distributed array system under the following settings:

- **Array Configuration:** The distributed array consists of $L = 4$ collinear ULAs. Each ULA is equipped with $M = 32$ sensors with an inter-element spacing of $d = 0.05$ m. The inter-subarray spacings are set to 5 m, 10 m, and 15 m, respectively.
- **Signal Parameters:** We assume three far-field wideband sources impinge on the array from DOAs $\{-50.85^\circ, -44.31^\circ, -38.36^\circ\}$. The sources have carrier frequencies of 3.002 GHz, 3.003 GHz, and 3.004 GHz, with corresponding bandwidths of 10 MHz, 30 MHz, and 15 MHz.
- **Baselines and Algorithms:** We compare the proposed SC-VBI and IC-VBI (which computes the posterior mean via direct matrix inversion) with three baseline methods: conventional beamforming (CBF) [12], SBL [13], and SPICE [14]. The baselines process each subarray separately without phase-error calibration, whereas JE-DPS jointly estimates the DOAs, phase errors, and sparse signals.

TABLE I
IMPACT OF THE NUMBER OF SUBARRAYS L AND SENSORS PER SUBARRAY M ON THE PROPOSED METHOD AT SNR = 0 dB.

Setting	Configuration	θ RMSE (rad)	φ RMSE (rad)
Varying L	$L = 2, M = 32$	0.00220	0.0640
	$L = 4, M = 32$	0.00127	0.0394
	$L = 6, M = 32$	0.00098	0.0305
Varying M	$L = 4, M = 16$	0.00195	0.0570
	$L = 4, M = 32$	0.00127	0.0394
	$L = 4, M = 48$	0.00105	0.0325

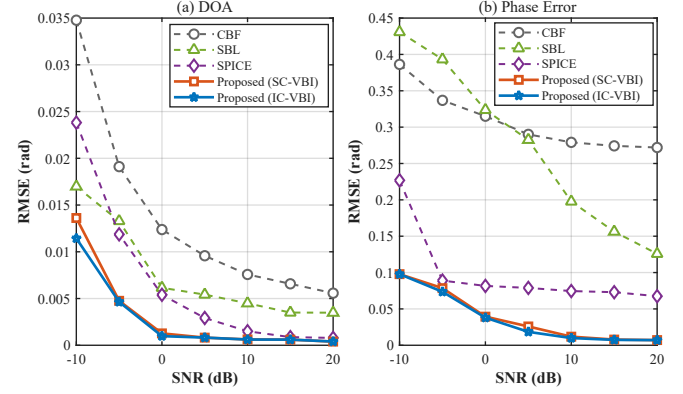


Fig. 4. RMSE versus SNR for different methods: (a) wideband DOA estimation; (b) phase-error estimation.

Fig. 4 shows the RMSE versus SNR for wideband DOA estimation in Fig. 4(a) and phase-error estimation in Fig. 4(b). All methods improve as the SNR increases. However, CBF and SBL exhibit inferior performance because they do not exploit the coherent gain provided by the distributed array. In contrast, the proposed methods consistently outperform the baselines across the tested SNR range, and the low-complexity SC-VBI achieves performance close to that of the high-complexity IC-VBI. The proposed method consistently outperforms the baselines, which further validates the effectiveness of the EM-based joint refinement.

Table I further illustrates the impact of the array configuration on the proposed method. As expected, increasing L improves the coherent gain across subarrays, while increasing M enhances the aperture of each subarray, and both lead to improved DOA and phase-error estimation accuracy.

VI. CONCLUSIONS

We proposed a joint wideband DOA and phase-error estimation algorithm for distributed array systems with unknown phase errors. By integrating SC-VBI into an EM framework, the proposed JE-DPS method effectively restores the coherent gain of distributed arrays and achieves high-resolution estimation with reduced computational complexity. Simulation results demonstrate the effectiveness of the proposed method in both DOA and phase-error estimation.

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